COURSE

IB Diploma Programme Business management HL

Stock

Unit 5: Operations management · 5.6 Production planning (HL only)

Revision for Unit 5: Operations management: each firm works it out on its own deals.

Class **IB Diploma Programme Business management HL** Topic **Stock**

THE LESSON

Setup	2 min

Round 1 — Decisions 3 min · Deals 5 min · Topic tasks 4 min

Firm decisions	**Deals**	TASK 1 **Stock holding cost**	TASK 2 **Risks of JIT**

Round 2 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 3 **Rank stock turnover**	TASK 4 **Stock turnover**	**B2B deal**

Round 3 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 5 **Cut or raise buffer**	TASK 6 **Weeks until reorder**	**B2B deal**

Final questions	3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

Cars wait in your yard.

Cost of holding the cars?

WORK OUT THE NUMBER · EXAMPLE FIGURES

You hold for them	**10 cars**
Holding cost per car a week	**£90**
Weeks you hold them	**6 weeks**

METHOD

Holding cost = cars × cost per car a week × weeks

10 × £90 × 6 = £5,400

Why: Stock that waits costs £900 every week.

Task 2

ROUND 1

You move to just-in-time stock.

Which two risks rise most?

PICK TWO

- ✔ **A late delivery stops sales**
- ✔ **Fewer bulk-buying discounts**
- ○ Higher storage costs
- ○ More cars go out of date
- ○ More cash tied up in stock

Why: JIT frees cash and space, but depends on every delivery and buys in small lots.

Task 3

ROUND 2

Three dealers: cost of cars sold in a year, and average stock.

Order by stock turnover, fastest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£450,000 sold, £50,000 stock**
2. **£600,000 sold, £100,000 stock**
3. **£800,000 sold, £200,000 stock**

Why: Turnover: 9, 6 and 4 times a year. The biggest dealer is the slowest.

Task 4

ROUND 2

One model, one year.

Stock turnover, times a year? Use average stock.

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Cost of cars sold in a year	**£167,400**
Stock at start of year	**£15,420**
Stock at end of year	**£21,780**

METHOD

Turnover = cost of cars sold ÷ average stock

(£15,420 + £21,780) ÷ 2 = £18,600; £167,400 ÷ £18,600 = 9

Why: The stock is sold and replaced 9 times a year.

Task 5

ROUND 3

Deliveries now come a week late. A lost sale costs £1,500; one more car in stock costs £80 a week.

What should you do with buffer stock, and why?

PICK A OR B AND GIVE A REASON

- ○ Cut it to save cash
- ✔ **Raise it**

Why: A lost sale costs far more than a week's holding cost, so a bigger buffer pays.

Task 6

ROUND 3

When do you order next?

Weeks until you reorder?

WORK OUT THE NUMBER · EXAMPLE FIGURES

They buy each week	**4 cars**
Supplier delivers in	**2 weeks**
Buffer, of lead-time sales	**50%**
In stock	**28 cars**

METHOD

Weeks = (stock – reorder level) ÷ weekly sales; buffer = half of lead-time sales

Lead-time sales 8, buffer 4, level 12; (28 − 12) ÷ 4 = 4

Why: A buffer tied to lead time grows when deliveries are slow.

Final questions

6	109	5
questions for each pupil	Level 3 Business questions	of them on stock

The game draws the 5 questions on this topic, and 1 more from the 109 Level 3 questions.

1 What is the central trade-off when a dealership adopts just-in-time delivery for common parts?
Lower holding cost, more supply risk.

2 A critical component supplier fails with no warning. Which earlier policy would most likely have reduced impact without necessarily holding maximum stock?
Tested backup suppliers for critical parts.

3 The cheapest supplier delivers only 72% of orders on time and in full, causing cancellations. What should Operations do?
Weigh failure cost against price.

4 A workshop uses just-in-time parts supply. Its main supplier shuts for two weeks after a fire. What is the most likely effect?
No buffer, so jobs stop at once.

5 Your only parts supplier’s lead time rises from two days to nine. Apart from ordering earlier, what else must change?
More stock must be held, tying up cash and space.